Name: Debarshi Das Adhikary
Roll No: 255hsbb011

Alternative Splicing Analysis of *Saccharomyces cerevisiae* RNA-Seq Data using HISAT2, SAMtools and SGSeq

1. Aim

To align single-end RNA-Seq reads from two conditions (wild type, WT, and mutant, MT; two replicates each) to the yeast reference genome, and to detect and visualise splice junctions and exons with the Bioconductor package SGSeq, in order to check whether the gene YAL030W shows any alternative splicing.

2. Data and Tools

Reference genome: GCF_000146045.2_R64_genomic.fna; annotation: GCF_000146045.2_R64_genomic.gtf (*S. cerevisiae* R64). Reads: conA_rep1.fq, conA_rep2.fq (WT1, WT2) and conB_rep1.fq, conB_rep2.fq (MT1, MT2), all single-end. Tools: HISAT2 (spliced alignment), SAMtools (SAM/BAM handling), and R/RStudio with SGSeq (splice graph analysis).

3. Methods

Part A: Terminal (alignment)

Step 1: Create and activate the conda environment

```
conda create -n altsplice -c conda-forge -c bioconda hisat2 samtools -y
conda activate altsplice
```

Step 2: Check the installed versions

```
hisat2 --version
samtools --version
```

The working folder contains the four read files, the genome FASTA and the GTF annotation (Figure 1).

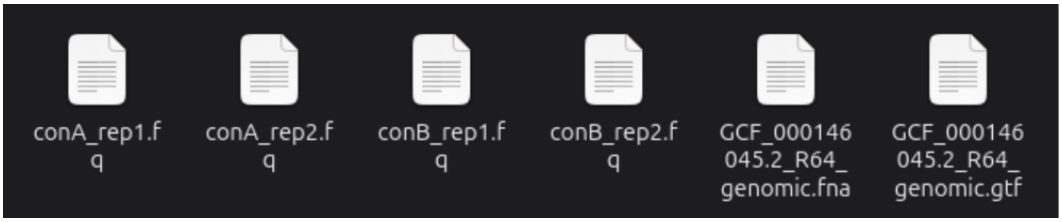


Figure 1. Input files in the working folder.

Step 3: Index the reference genome

```
hisat2-build GCF_000146045.2_R64_genomic.fna Hisat_index
ls Hisat_index*
```

Step 4: Align the single-end reads

```
hisat2 -x Hisat_index -U conA_rep1.fq -S A_1.sam
hisat2 -x Hisat_index -U conA_rep2.fq -S A_2.sam
hisat2 -x Hisat_index -U conB_rep1.fq -S B_1.sam
hisat2 -x Hisat_index -U conB_rep2.fq -S B_2.sam
```

Alignment summary reported by HISAT2 (for the sample shown):

```
800000 reads; of these:
  800000 (100.00%) were unpaired; of these:
    31586 (3.95%) aligned 0 times
    717191 (89.65%) aligned exactly 1 time
    51223 (6.40%) aligned >1 times
96.05% overall alignment rate
```

Step 5: Convert SAM to BAM

```
samtools view -bS A_1.sam -o A_1.bam
samtools view -bS A_2.sam -o A_2.bam
samtools view -bS B_1.sam -o B_1.bam
samtools view -bS B_2.sam -o B_2.bam
```

Step 6: Sort the BAM files

```
samtools sort -o A_1_sorted.bam A_1.bam
samtools sort -o A_2_sorted.bam A_2.bam
samtools sort -o B_1_sorted.bam B_1.bam
samtools sort -o B_2_sorted.bam B_2.bam
```

Step 7: Index the sorted BAM files

```
samtools index A_1_sorted.bam
samtools index A_2_sorted.bam
samtools index B_1_sorted.bam
samtools index B_2_sorted.bam
ls *_sorted.bam*
```

Part B: RStudio (SGSeq splice graph analysis)

In RStudio, the working directory is set to the folder containing the sorted BAM files and the annotation.

Step 8: Install and load SGSeq (install only the first time)

```
setwd("~/bio-env")
if (!requireNamespace("BiocManager", quietly = TRUE)) install.packages("BiocManager")
BiocManager::install("SGSeq")
library(SGSeq)
```

Step 9: Prepare the input data

The column names must be exactly *sample_name* and *file_bam*.

```
file_bam <- c("A_1_sorted.bam", "A_2_sorted.bam", "B_1_sorted.bam", "B_2_sorted.bam")
sample_name <- c("WT1", "WT2", "MT1", "MT2")
v <- data.frame(sample_name, file_bam)
v
```

```
  sample_name       file_bam
1         WT1 A_1_sorted.bam
2         WT2 A_2_sorted.bam
3         MT1 B_1_sorted.bam
4         MT2 B_2_sorted.bam
```

Step 10: Extract BAM file information

```
bam_info <- getBamInfo(v)
bam_info
```

```
  sample_name       file_bam paired_end read_length frag_length lib_size
1         WT1 A_1_sorted.bam      FALSE         101          NA   768414
2         WT2 A_2_sorted.bam      FALSE         101          NA   867936
3         MT1 B_1_sorted.bam      FALSE         101          NA   673580
4         MT2 B_2_sorted.bam      FALSE         101          NA   961359
```

Step 11: Import the annotation

```
transcripts <- importTranscripts(file = "GCF_000146045.2_R64_genomic.gtf")
tx_features <- convertToTxFeatures(transcripts)
```

Step 12: Predict features and quantify expression

```
AnaFeat <- analyzeFeatures(bam_info, features = tx_features)
FPKM(AnaFeat)
rowRanges(AnaFeat)
```

```
MT1 complete.
MT2 complete.
                 WT1           WT2          MT1          MT2
   [1,] 0.000000e+00 0.000000e+00    0.0000000    0.0000000
   [2,] 0.000000e+00 2.395340e+00    3.0864962    0.0000000
   [3,] 0.000000e+00 0.000000e+00    0.0000000    0.0000000
   [4,] 2.913541e+01 1.328813e+01   13.0935281    9.8797273
   [5,] 6.189733e+02 3.233532e+02  294.9473049  174.3900075
   [6,] 7.897898e+02 7.850176e+02  484.9629280  410.5810344
   [7,] 1.408803e+01 1.870894e+01   20.0893728   43.6346663
   [8,] 2.830849e+01 1.861785e+01   31.3713848   30.3847885
   [9,] 6.357461e+01 3.780324e+01   28.6853979   37.5425520
  [10,] 3.035883e+01 2.687773e+01   16.1621035   14.5594825
  [11,] 7.908071e+01 7.565153e+01   81.7380211   73.3905328
  [12,] 2.647694e+01 2.308579e+01   25.6281937   29.1793055
  [13,] 9.016502e+01 4.789574e+01   48.0010741   50.4482147
  [14,] 1.187463e+02 1.538491e+02  158.5927816  171.3079373
  [15,] 3.271767e+01 2.464279e+01   28.9678956   40.9832589
  [16,] 2.910452e+03 4.724354e+03 3482.2916302 3550.8062034
```

Figure 2. Output of FPKM(AnaFeat): FPKM values of each feature (rows) in WT1, WT2, MT1 and MT2 (columns).

```
SGFeatures object with 7787 ranges and 0 metadata columns:
           seqnames      ranges strand     type  splice5p  splice3p featureID
              <Rle>   <IRanges>  <Rle> <factor> <logical> <logical> <integer>
    [1] NC_001133.9   2480-2707      +        E     FALSE     FALSE         1
    [2] NC_001133.9 12046-12426      +        E     FALSE     FALSE         2
    [3] NC_001133.9 21566-21850      +        E     FALSE     FALSE         3
    [4] NC_001133.9 31567-32940      +        E     FALSE     FALSE         4
    [5] NC_001133.9 33448-34701      +        E     FALSE     FALSE         5
    ...         ...          ...    ...      ...       ...       ...       ...
 [7783] NC_001224.1 78533-78608      +        E     FALSE     FALSE      7783
 [7784] NC_001224.1 79213-80022      +        E     FALSE     FALSE      7784
 [7785] NC_001224.1 85035-85112      +        E     FALSE     FALSE      7785
 [7786] NC_001224.1 85295-85777      +        E     FALSE     FALSE      7786
 [7787] NC_001224.1 78089-78162      -        E     FALSE     FALSE      7787
          geneID                txName        geneName
       <integer>        <CharacterList> <CharacterList>
    [1]         1         NM_001184582.1       YAL067W-A
    [2]         2         NM_001180042.1       YAL064W-B
    [3]         3         NM_001178206.2         YAL064W
    [4]         4         NM_001178204.1         YAL062W
    [5]         5         NM_001178203.1         YAL061W
    ...       ...                    ...             ...
 [7783]      6345 unassigned_transcrip..        tV(UAC)Q
 [7784]      6346 unassigned_transcrip..           Q0275
 [7785]      6347 unassigned_transcrip..       tM(CAU)Q2
 [7786]      6348 unassigned_transcrip..           Q0285
 [7787]      6349 unassigned_transcrip..       tT(UAG)Q2
 -------
 seqinfo: 17 sequences from an unspecified genome; no seqlengths
```

Figure 3. Output of rowRanges(AnaFeat): the SGFeatures object.

Step 13: Visualise splice features

```
plotFeatures(AnaFeat, geneName = "YAL030W")
```

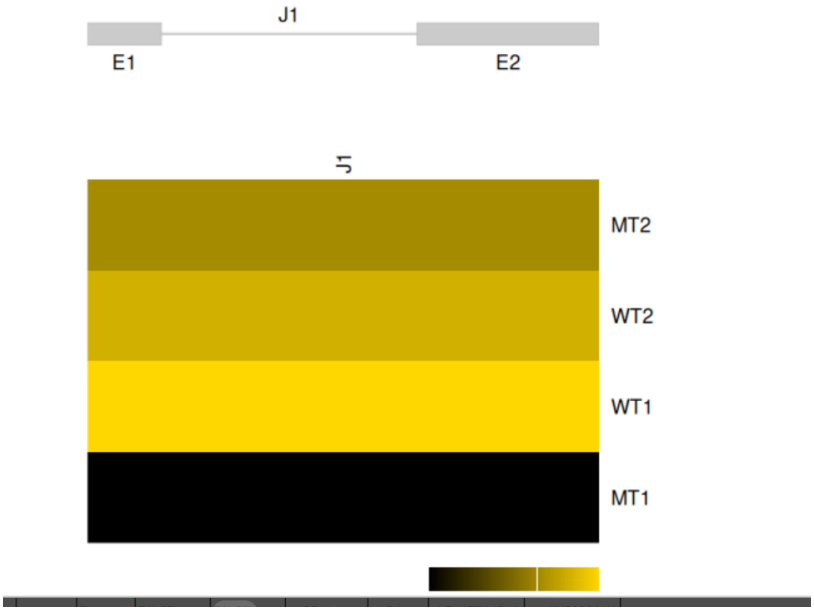


Figure 4. plotFeatures() for YAL030W (default call).

For a better plot, a modified version of the function was used. It matches features by featureID first, then by name, and drops unmatched (NA) entries, which avoids the indexing problem of the default function.

```
plotFeatures_fixed <- function(x, geneID = NULL, geneName = NULL, which = NULL,
    tx_view = FALSE, cex = 1, assay = "FPKM",
    include = c("junctions", "exons", "both"),
    transform = function(x) log2(x + 1),
    Rowv = NULL, distfun = dist, hclustfun = hclust, margin = 0.2,
    RowSideColors = NULL, square = FALSE, cexRow = 1, cexCol = 1,
    labRow = colnames(x),
    col = colorRampPalette(c("black", "gold"))(256),
    zlim = NULL, heightPanels = c(1, 2), ...)
{
    include <- match.arg(include)
    x <- restrictFeatures(x, geneID = geneID, which = which, geneName = geneName)
    if (nrow(x) == 0) return()
    features <- rowRanges(x)
    if (!is.null(RowSideColors) && !is.list(RowSideColors)) {
        RowSideColors <- list(RowSideColors)
    }
    n_sample <- ncol(x)
    include_type <- switch(include, junctions = "J", exons = "E", both = c("E", "J"))
    n_feature <- length(which(type(features) %in% include_type))
    pars <- getLayoutParameters(n_sample, n_feature, margin, heightPanels,
                                RowSideColors, square, tx_view)
    layout(pars$mat, heights = pars$hei, widths = pars$wid)
    par(cex = cex, mai = c(pars$mai[1], 0, pars$mai[3], 0))
    df <- plotSpliceGraph(x = features, label = "id", tx_view = tx_view,
                          short_output = FALSE, ...)
    i_df <- switch(include,
                   exons = which(df$type == "E"),
                   junctions = which(df$type == "J"),
                   both = seq_len(nrow(df)))
    i_df <- i_df[!duplicated(df$name[i_df])]
    # FIX: match on featureID first, then fall back to name, then drop NAs
    i <- match(df$featureID[i_df], featureID(features))
    miss <- is.na(i)
    if (any(miss)) {
        i[miss] <- match(df$name[i_df][miss], feature2name(features))
    }
    keep <- !is.na(i)
```

```
    i_df <- i_df[keep]
    i <- i[keep]
    X <- transform(assay(x[i, ], assay))
    labCol <- df$id[i_df]
    colLabCol <- df$color[i_df]
    plotImage(X, Rowv = Rowv, distfun = distfun, hclustfun = hclustfun,
              RowSideColors = RowSideColors, cexRow = cexRow, cexCol = cexCol,
              labRow = labRow, labCol = labCol, colLabCol = colLabCol,
              col = col, zlim = zlim)
    frame()
    df <- df[c("id", "name", "type", "featureID")]
    invisible(df)
}

# let it see SGSeq's internal functions (getLayoutParameters, plotImage, etc.)
environment(plotFeatures_fixed) <- asNamespace("SGSeq")

while (dev.cur() > 1) dev.off()
plotFeatures_fixed(AnaFeat, geneID = 21, include = "both")
```

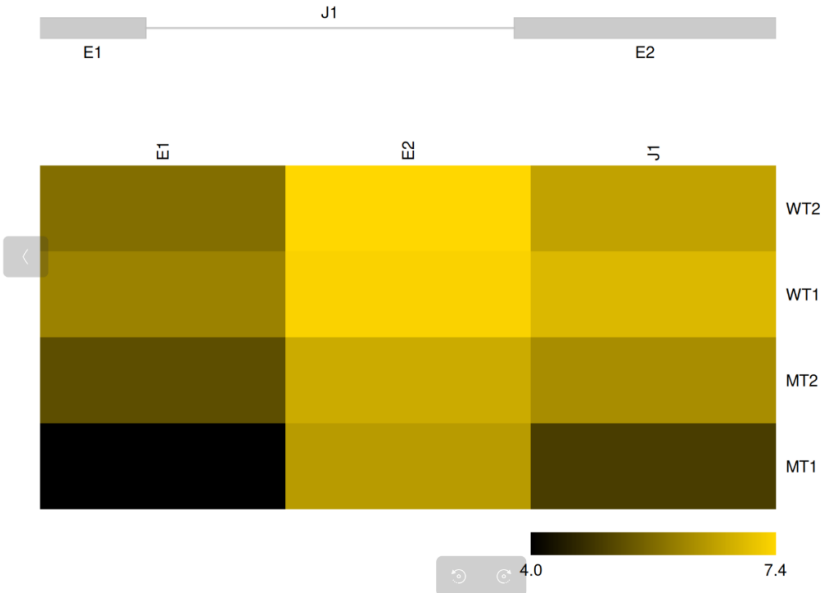


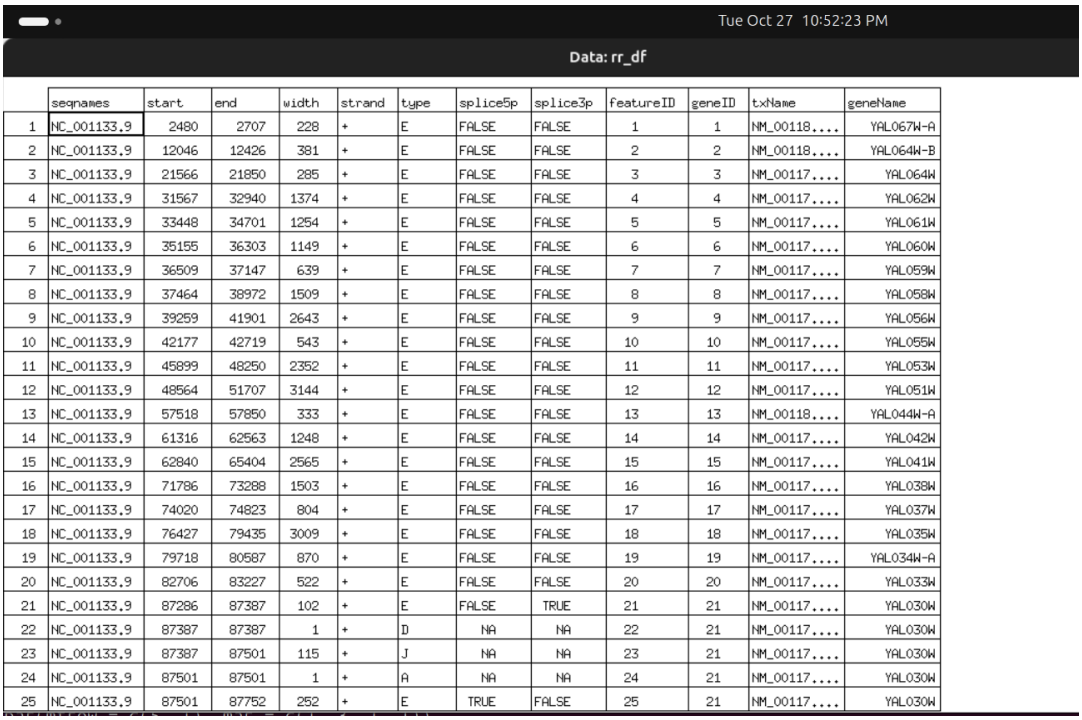
Figure 5. plotFeatures_fixed() for gene ID 21 (YAL030W): exons E1, E2 and junction J1, log2(FPKM + 1).

Step 14: Find the gene ID, then plot the splice graph and coverage

```
rr_df <- as.data.frame(rowRanges(AnaFeat))
View(rr_df)    # find the geneID of your gene of interest (e.g. YAL030W)

par(mfrow = c(5, 1), mar = c(1, 3, 1, 1))
plotSpliceGraph(rowRanges(AnaFeat), geneID = 505, toscale = "none", color_novel = "red")
for (j in 1:4) {
  plotCoverage(AnaFeat[, j], geneID = 505, toscale = "none")
}

# save the figure as a PNG file
png("splicegraph_coverage.png", width = 1800, height = 2000, res = 200)
par(mfrow = c(5, 1), mar = c(1, 3, 1, 1))
plotSpliceGraph(rowRanges(AnaFeat), geneID = 505, toscale = "none", color_novel = "red")
for (j in 1:4) plotCoverage(AnaFeat[, j], geneID = 505, toscale = "none")
dev.off()
```



	seqnames	start	end	width	strand	type	splice5p	splice3p	featureID	geneID	txName	geneName
1	NC_001133.9	2480	2707	228	+	E	FALSE	FALSE	1	1	NM_00118....	YAL067W-A
2	NC_001133.9	12046	12426	381	+	E	FALSE	FALSE	2	2	NM_00118....	YAL064W-B
3	NC_001133.9	21566	21850	285	+	E	FALSE	FALSE	3	3	NM_00117....	YAL064W
4	NC_001133.9	31567	32940	1374	+	E	FALSE	FALSE	4	4	NM_00117....	YAL062W
5	NC_001133.9	33448	34701	1254	+	E	FALSE	FALSE	5	5	NM_00117....	YAL061W
6	NC_001133.9	35155	36303	1149	+	E	FALSE	FALSE	6	6	NM_00117....	YAL060W
7	NC_001133.9	36509	37147	639	+	E	FALSE	FALSE	7	7	NM_00117....	YAL059W
8	NC_001133.9	37464	38972	1509	+	E	FALSE	FALSE	8	8	NM_00117....	YAL058W
9	NC_001133.9	39259	41901	2643	+	E	FALSE	FALSE	9	9	NM_00117....	YAL056W
10	NC_001133.9	42177	42719	543	+	E	FALSE	FALSE	10	10	NM_00117....	YAL055W
11	NC_001133.9	45899	48250	2352	+	E	FALSE	FALSE	11	11	NM_00117....	YAL053W
12	NC_001133.9	48564	51707	3144	+	E	FALSE	FALSE	12	12	NM_00117....	YAL051W
13	NC_001133.9	57518	57850	333	+	E	FALSE	FALSE	13	13	NM_00118....	YAL044W-A
14	NC_001133.9	61316	62563	1248	+	E	FALSE	FALSE	14	14	NM_00117....	YAL042W
15	NC_001133.9	62840	65404	2565	+	E	FALSE	FALSE	15	15	NM_00117....	YAL041W
16	NC_001133.9	71786	73288	1503	+	E	FALSE	FALSE	16	16	NM_00117....	YAL038W
17	NC_001133.9	74020	74823	804	+	E	FALSE	FALSE	17	17	NM_00117....	YAL037W
18	NC_001133.9	76427	79435	3009	+	E	FALSE	FALSE	18	18	NM_00117....	YAL035W
19	NC_001133.9	79718	80587	870	+	E	FALSE	FALSE	19	19	NM_00117....	YAL034W-A
20	NC_001133.9	82706	83227	522	+	E	FALSE	FALSE	20	20	NM_00117....	YAL033W
21	NC_001133.9	87286	87387	102	+	E	FALSE	TRUE	21	21	NM_00117....	YAL030W
22	NC_001133.9	87387	87387	1	+	D	NA	NA	22	21	NM_00117....	YAL030W
23	NC_001133.9	87387	87501	115	+	J	NA	NA	23	21	NM_00117....	YAL030W
24	NC_001133.9	87501	87501	1	+	A	NA	NA	24	21	NM_00117....	YAL030W
25	NC_001133.9	87501	87752	252	+	E	TRUE	FALSE	25	21	NM_00117....	YAL030W

Figure 6. rr_df table (rowRanges as a data frame) with the features of YAL030W (gene ID 21).

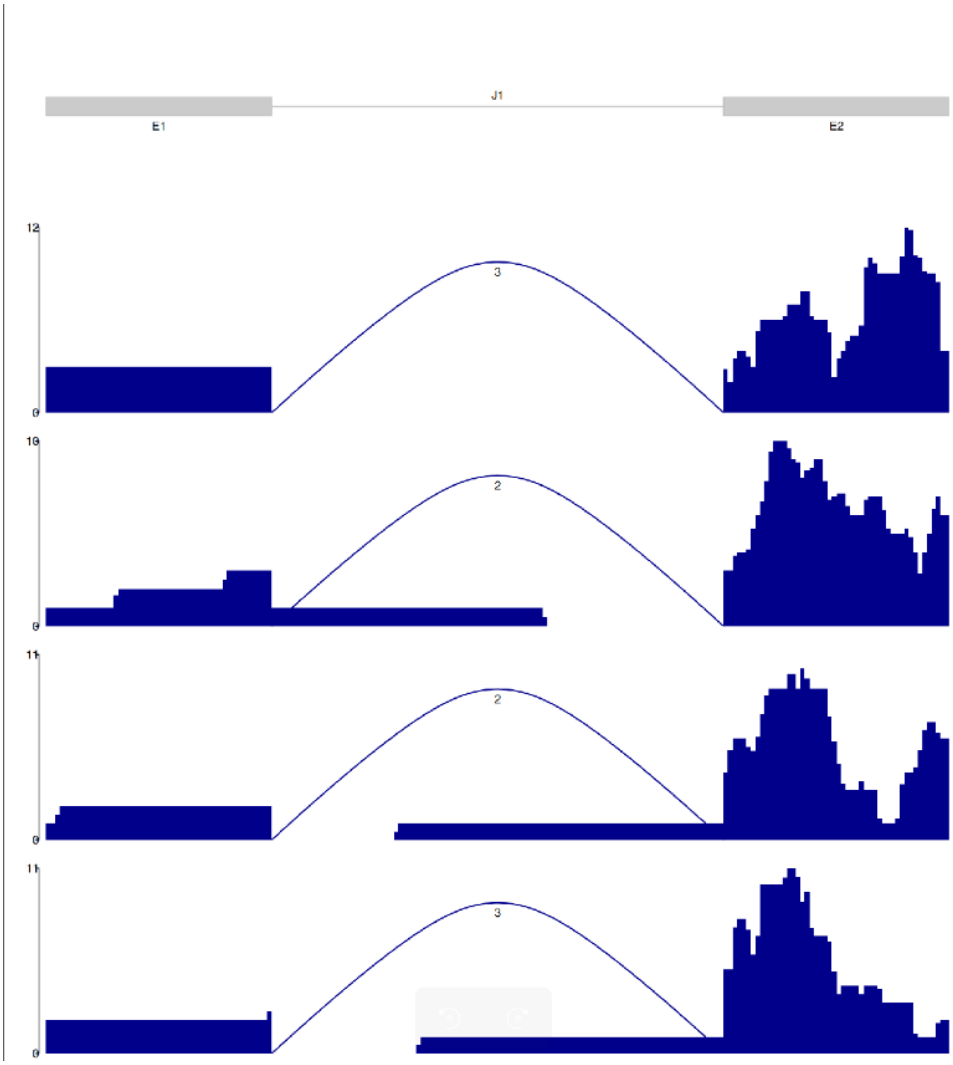


Figure 7. Splice graph (top) and read coverage of WT1, WT2, MT1 and MT2 (panels 2 to 5, top to bottom).

4. Results and Interpretation

4.1 Alignment quality

Out of 800,000 single-end reads, 96.05% aligned to the reference genome. 89.65% (717,191 reads) aligned uniquely and 6.40% (51,223 reads) aligned to more than one location, while only 3.95% (31,586 reads) did not align. A high overall alignment rate with mostly unique hits shows that the reads are of good quality, the correct reference was used, and the data are suitable for splice junction analysis.

4.2 BAM information and library sizes

All four samples are single-end (paired_end = FALSE) with a read length of 101 bp, so the fragment length is NA. The library sizes differ: WT1 = 768,414, WT2 = 867,936, MT1 = 673,580 and MT2 = 961,359 reads. MT1 is the smallest and MT2 the largest library (about 1.4 times larger), so raw read counts cannot be compared directly between samples. This is why expression was compared as FPKM, which normalises for both library size and feature length.

4.3 Annotation and features (Figure 3)

The SGFeatures object contains 7,787 ranges. The first entries are exons (type E) on NC_001133.9 (chromosome I) with gene names such as YAL067W-A, YAL064W-B and YAL064W, and the last entries belong to non-coding genes (tRNAs and others, for example tV(UAC)Q and Q0275) on the mitochondrial sequence NC_001224.1. The columns splice5p and splice3p are FALSE for most exons, which means that most annotated exons are not flanked by a splice site, as expected in yeast, where the majority of genes have no introns. The note "17 sequences from an unspecified genome; no seqlengths" only means that chromosome lengths were not stored in the object; it is not an error.

4.4 FPKM values (Figure 2)

The FPKM table has one row per feature and one column per sample. Some features (rows 1 and 3) have FPKM of 0 in all samples, which means they are not expressed or have no reads. Other features are expressed at very different levels, from about 10 to several thousand FPKM (for example row 6: 789.8, 785.0, 485.0 and 410.6 in WT1, WT2, MT1 and MT2; row 16 reaches the thousands). In many rows the two WT samples have higher values than the two MT samples (rows 5, 6 and 7 are clear examples), but the two replicates of the same condition are close to each other, which indicates good reproducibility. The reverse is also seen in a few rows (for example rows 14 and 15 show higher values in MT), so the changes are feature specific and not a general shift.

4.5 Gene YAL030W: exons and junction (Figures 4, 5 and 6)

The rr_df table shows that gene ID 21 (YAL030W) has five features: an exon at 87286 to 87387 (102 bp, splice site at its end), a splice donor (type D) at 87387, a junction (type J) from 87387 to 87501 (115 bp), a splice acceptor (type A) at 87501, and a second exon at 87501 to 87752 (252 bp, splice site at its start). In the plots these are named E1, J1 and E2. So YAL030W has only one annotated splicing event: E1 is joined to E2 by J1, and the intron lies between them. No alternative exon or alternative junction is present in the splice graph.

Figure 4 (default plotFeatures) shows only the junction J1 as a single heat map column. J1 is brightest in WT1, slightly less bright in WT2 and MT2, and darkest (black) in MT1. That means the splicing junction is supported by the most reads in WT1 and by the fewest reads in MT1. A single column is difficult to compare, so the modified function was used.

Figure 5 (modified function, both exons and junction) shows E1, E2 and J1 as columns and the four samples as rows. The colour scale is log2(FPKM + 1) and runs from 4.0 (black, low) to 7.4 (gold, high). Exon E2 is bright in all four samples, so it is the most highly expressed part of the gene, and it is similar in WT1, WT2 and MT2 and a little lower in MT1. Exon E1 is darker than E2 in every sample and is nearly black in MT1, so MT1 has the lowest expression of E1. The junction J1 has intermediate intensity: higher in WT1 and WT2 and lower in MT2 and MT1. Overall, the WT samples show a little higher expression of E1 and J1 than the MT samples, and MT1 is the lowest in all features. Since MT1 also has the smallest library, part of this difference may be a sampling effect, and with only two replicates per condition the difference cannot be called statistically significant.

4.6 Splice graph and coverage (Figure 7)

The top panel is the splice graph of the gene: two exons (E1 and E2, grey boxes) connected by one junction J1 (line). Novel features would be drawn in red because of color_novel = "red", but no red feature is present, so no novel splice site or novel exon was found. The four lower panels show the read coverage of WT1, WT2, MT1 and MT2. The arc over the intron represents reads that span the junction, and the number on the arc is the junction read count: 3 reads in WT1, 2 in WT2, 2 in MT1 and 3 in MT2. The maximum coverage on the y axis is about 12 for WT1, 19 for WT2 and 11 for MT1 and MT2.

In all four samples the coverage is low and almost flat in E1 (about 2 to 3 reads) and much higher and uneven in E2, with peaks up to 12 to 19 reads. This agrees with the heat map, where E2 was brighter than E1. In the intron region there is almost no coverage in WT1, while a thin line of about 1 read appears in WT2, MT1 and MT2. This low-level signal may come from a few unspliced or partially processed transcripts or from background, but it is far too small to indicate intron retention.

All four samples have the same pattern, that is, E1 spliced to E2 through J1, with the same junction. The differences between WT and MT are in the amount of reads, not in the way the gene is spliced.

5. Conclusion

The reads aligned well to the *S. cerevisiae* genome (96.05% overall alignment). SGSeq identified 7,787 features in the annotation and quantified them as FPKM in the WT and MT samples. For the gene YAL030W, one splice junction (J1) connecting two exons (E1 and E2) was found, and it is present in all four samples with 2 to 3 junction-spanning reads each. No alternative splice isoform and no novel splicing event was detected for this gene. E2 is the most highly covered region, and WT samples show slightly higher E1 and J1 expression than MT samples, but this is based on few reads and two replicates only, so a statistical test with more replicates would be needed to confirm any difference.